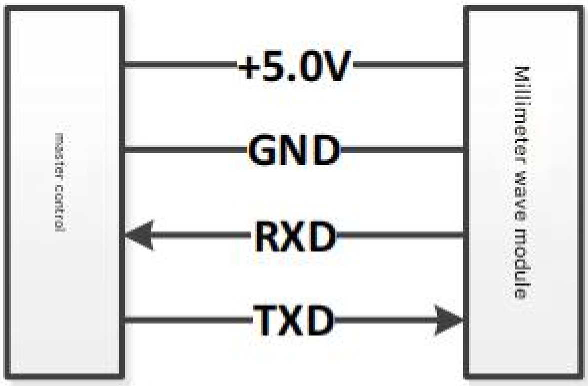
# Intelligent Air Conditioning Radar LD6001

## Protocol Description Document V1.1

# 1. Hardware Connection

Using UART communication, the hardware connection is as follows (this figure only shows the UART wiring, the actual line sequence is subject to the hardware design requirements):




# 2. Communication parameters

Baud rate: 9600

Data bits: 8

Stop bits: 1

Checksum: Even

Flow Control: None

# 3. Message Definition

### 3.1 Message Format

| | |
|---|---|
| **Byte0** | **0x44** from master to module , **0x4D** from module to master |

| | |
|---|---|
| **Byte1** | Message ID, see the definition of each message |
| **Byte2** | Data length (Byte4~Byte(N-3))<br>-Must be a multiple of 8<br>-Length range 0~248<br>- The request and response message lengths can be different |
| **Byte3** | Reserved, default value is 00 |
| **Byte4** | Data0 |
| **Byte5** | Data 1 |
| **...** | ... |
| **ByteN-2** | CheckSum, add from Byte0 to Byte(N-3) |
| **ByteN-1** | **0x4B** from master to module , **0x4A from module to master** |

### 3.2 Query module status information 0x11

| |
|---|
| The master query module version information. |

| | |
|---|---|
| Byte0 | 0x44 |
| Byte1 | 0x11 |
| Byte2 | 0x00 |
| Byte3 | 0x00 |
| Byte4 | Checksum |
| Byte5 | 0x4B |
| Module response | |

| | |
|---|---|
| Byte0 | 0x4D |
| Byte1 | 0x11 |
| Byte2 | 0x08 |
| Byte3 | 0x00 |
| Byte4 | Software minor version number |
| Byte5 | Software major version number |
| Byte6 | Hardware minor version number |
| Byte7 | Hardware major version number |
| Byte8 | Reserve |
| Byte9 | Current working status: 0: Initialization completed; 1: Initializing |
| Byte10 | Reserve |
| Byte11 | Reserve |
| Byte12 | Checksum |
| Byte13 | 0x4A |

### 3.3 Get radar detection data command 0x62

| The detection data of the master query module | |
|---|---|
| Byte0 | 0x44 |
| Byte1 | 0x62 |
| Byte2 | 0x08 |
| Byte3 | 0x00 |
| Byte4 | Sensitivity attribute, 0x10 normal sensitivity, 0x20 high sensitivity |
| Byte5 | 0x00 |
| Byte6 | 0x00 |
| Byte7 | 0x00 |
| Byte8 | 0x00 |
| Byte9 | 0x00 |
| Byte10 | 0x00 |
| Byte11 | 0x00 |
| Byte12 | Checksum |
| Byte13 | 0x4B |

<table>
<tr><td colspan="3">Module response</td></tr>
<tr><td>Byte0</td><td colspan="2">0x4D</td></tr>
<tr><td>Byte1</td><td colspan="2">0x62</td></tr>
<tr><td>Byte2</td><td colspan="2">Length: number of targets (M+1)*8, the maximum value</td></tr>
<tr><td>Byte3</td><td colspan="2">0x00</td></tr>
<tr><td>Byte4</td><td colspan="2">Millimeter wave module fault status, 0 means no fault,</td></tr>
<tr><td>Byte5</td><td colspan="2">The number of detected targets is M (the maximum value</td></tr>
<tr><td>Byte6</td><td colspan="2">Reserve</td></tr>
<tr><td>Byte7</td><td colspan="2">Reserve</td></tr>
<tr><td>Byte8</td><td colspan="2">Reserve</td></tr>
<tr><td>Byte9</td><td colspan="2">Reserve</td></tr>
<tr><td>Byte10</td><td colspan="2">Reserve</td></tr>
<tr><td>Byte11</td><td colspan="2">Reserve</td></tr>
<tr><td rowspan="4">Goal 1</td><td>Byte1</td><td>Target ID</td></tr>
<tr><td>Byte1</td><td>Target distance d (0.0-25.5m), unsigned uchar</td></tr>
<tr><td>Byte1</td><td>Target pitch angle θ (0~180 degrees), unsigned</td></tr>
<tr><td>Byte1</td><td>Target horizontal angle ∂ (0~180 degrees),</td></tr>
</table>

| | | |
|---|---|---|
| | Byte1 | Reserve |
| | Byte1 | Reserve |
| | Byte1 | Target X coordinate value, signed char type, unit |
| | Byte1 | Target Y coordinate value, signed char type, unit |
| Goal 2 | Byte2 | Data format is the same as above |
| … | … | … |
| Target | Byte( | Data format is the same as target 1 |
| ByteN-2 | Checksum | |
| ByteN-1 | 0x4A | |